Gustavo A. **Salazar-Gomez**

AI Research Scientist

gsg213.github.io | gustavo-salazar-gomez | github.com/gsg213

Profile

AI Research Scientist with 4+ years of deep learning research at Inria, designing Transformer and attention-based architectures for large-scale multi-modal and temporal data (camera, LiDAR, occupancy grids). Experienced in PyTorch, multi-GPU distributed training and reproducible experimentation, with publications at IEEE IV and CVPR workshops. Certified in Generative AI with Large Language Models, and motivated to apply these skills to biomedical and healthcare challenges.

Experience

Centre Inria de l'Université Grenoble Alpes — *Grenoble, France*

PhD Researcher - CHROMA team — *Jan 2024 - Dec 2026*

- Designed attention-based deep learning architectures (Transformer, ViT) for spatial reasoning over multiple-channel image inputs, processing temporal sequences of occupancy grids to produce accurate and safe trajectory outputs, creatively rethinking how scene information is represented to the network; trained them with multi-GPU distributed training on the Grid'5000 cluster.
- Analysed model behaviour through systematic ablation studies of input representations, architecture choices (EfficientNet, ResNet, ViT, Transformer) and loss configurations, breaking open-ended research questions into focused, testable experiments kept reproducible with Hydra configurations and Weights & Biases / TensorBoard tracking.
- Bridged perception, motion planning and safety perspectives within an interdisciplinary robotics team, and communicated findings through scientific publications and publicly released code.

Centre Inria de l'Université Grenoble Alpes — *Grenoble, France*

R&D Engineer - CHROMA team — *Oct 2022 - Dec 2023*

- Carried out research on Transformer-based, attention-driven fusion architectures for dense semantic segmentation from multi-modal inputs (camera and LiDAR), with emphasis on temporal sequence modeling and robustness to occlusions on large datasets such as nuScenes, quickly adapting from an industry role to a research environment.
- Driven by curiosity about why fusion models fail under occlusion, explored new ideas beyond the initial scope and prepared results for presentation at related conferences, turning technical work into clear scientific communication.

Open Intelligence — *Cali, Colombia*

Product Specialist - Support Services — *Feb 2019 - Sept 2021*

- Coordinated technical and functional duties, configuring, debugging and supporting Open's product for North-American customers, working with SQL and PL/SQL on the product's databases, analysing reported issues to isolate their root cause and devising personalized functions to meet customer requirements.
- Established partnership and communication channels with customers to collect and understand their needs, acting as the bridge between business users and technical teams and adapting to each client's context to propose tailored solutions.

Robotica for kids — *Cali, Colombia*

Robotics teacher — *Jan 2018 - Dec 2018*

- Taught weekly robotics classes to middle and high-school students, explaining sensors, actuators and programming logic in accessible terms and adapting the pace to each group.
- Creatively designed hands-on projects at several complexity levels to spark students curiosity and keep classes engaging.

Education

PhD in Informatics — *Grenoble, France*

Université Grenoble Alpes | INRIA — *Jan 2024 - Dec 2026*

Under supervision of Anne Spalanzani and Christian Laugier
Subject: AI-driven algorithms for safe motion planning and driving decision-making in the context of autonomous driving. Defense expected February - Mars 2027.

M.Sc. in Mobile, Autonomous and Robotic Systems *Grenoble, France*

Grenoble INP | Université Grenoble Alpes *Sept 2021 - Sept 2022*

Dissertation: "Transformer-based Lidar-RGB Fusion for Semantic Grid Prediction in Autonomous Vehicles" - link

Postgraduate Diploma Specialization in Artificial Intelligence *Cali, Colombia*

Universidad Autonoma de Occidente (UAO) *Aug 2020 - Jul 2021*

B.Eng. in Mechatronics Engineering *Cali, Colombia*

Universidad Autonoma de Occidente (UAO) *Jan 2013 - Dec 2017*

Dissertation: "Object recognition in images using Deep Learning"

Technical Skills

- **Deep Learning:** Transformers & Attention Mechanisms, Vision Transformers (ViT), Multi-modal Data Fusion, Image Segmentation, Object Detection & Tracking, Temporal Sequence Modeling, Architecture Design, Ablation Studies.
- **ML Engineering:** Multi-GPU Distributed Training (PyTorch DDP / Lightning on the Grid'5000 cluster), Experiment Tracking (Weights & Biases, TensorBoard), Configuration Management (Hydra), AWS SageMaker, Docker.
- **Frameworks & Libraries:** PyTorch, PyTorch Lightning, Hugging Face Transformers, TensorFlow, Scikit-Learn, OpenCV, NumPy, SciPy, Matplotlib, Open3D.
- **Programming & Tools:** Python, SQL, PL/SQL, Shell, Git (GitHub, GitLab), Linux.

Relevant Courses & Certificates

- **Generative AI with Large Language Models** - link *Oct 2026*

  DeepLearning.AI & Amazon Web Services on Coursera. Transformer-based generative architectures and the full LLM project lifecycle: pre-training, instruction fine-tuning and parameter-efficient fine-tuning, alignment with reinforcement learning from human feedback, model evaluation, and optimization for deployment. LLM-powered applications with retrieval-augmented generation (RAG) and reasoning-and-acting agents (ReAct, LangChain). Hands-on labs in Python with Hugging Face (Transformers, PEFT) on AWS services such as Amazon SageMaker.

Languages

- **Spanish:** Mother-Language
- **English:** Advanced - C1
- **French:** Intermediate - B1/B2

Publications

[1] **G. Salazar-Gomez**, A. Spalanzani, L. Rummelhard, and C. Laugier, "Planflow: Local trajectory planning for autonomous driving using flow-guided occupancy grids", in *2026 IEEE Intelligent Vehicles Symposium (IV)*, 2026, pp. 997–1003.

[2] **G. Salazar-Gomez**, W. Liu, M. Diaz-Zapata, D. Sierra-Gonzalez, and C. Laugier, "Tlcfuse: Temporal multi-modality fusion towards occlusion-aware semantic segmentation", in *2024 IEEE Intelligent Vehicles Symposium (IV)*, 2024, pp. 2110–2116.

[3] G. Salazar-Gomez, D. S. González, M. Diaz-Zapata, A. Paigwar, W. Liu, Ö. Erkent, and C. Laugier, "Transfusegrid: Transformer-based lidar-rgb fusion for semantic grid prediction", in *17th International Conference on Control, Automation, Robotics and Vision, ICARCV 2022, Singapore, Singapore, December 11-13, 2022*, IEEE, 2022, pp. 268–273.

[4] G. A. Salazar-Gomez, M. A. Saavedra-Ruiz, and V. Romero-Cano, "High-level camera-lidar fusion for 3d object detection with machine learning", *Computer Vision and Pattern Recognition (CVPR) Conference: LatinX in AI (LXAI) Research Workshop*, 2021.

[5] J. A. López Sotelo, N. Díaz Salazar, and G. A. Salazar Gomez, "Application of transfer learning for object recognition using convolutional neural networks", in *Applications of Computational Intelligence*, A. D. Orjuela-Cañón, J. C. Figueroa-García, and J. D. Arias-Londoño, Eds., Cham: Springer International Publishing, 2018, pp. 14–25, ISBN: 978-3-030-03023-0_2.